Routes to Pressure for Path Integral on Riemannian Manifolds

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We write the three pressure estimators for a quantum fluid on a Riemannian manifold: virial, thermodynamic, and kinetic. And we discuss how the virial estimator plays a privileged role among the three. When possible we describe the consistency between the three routes.

Keywords: Virial Pressure; Thermodynamic Pressure; Kinetic Pressure; Path Integral; Riemannian Manifold

CONTENTS

I. Introduction 1

II. Canonical variables 2

III. Primitive approximation for the density matrix 3

IV. Routes to the pressure 4
A. Virial route 4
B. Thermodynamic pressure 5
C. Kinetic pressure 7

V. Conclusions 7

Author declarations 7
Conflicts of interest 7
Data availability 8
Funding 8

References 8

I. INTRODUCTION

The description and the properties of a quantum fluid on a Riemannian manifold is a field still under fervid development [1, 2]. Guided by the well settled description of a fluid in Euclidean space [3–5] the extension to a curved space sees the appearance of a new actor represented by the metric tensor g_{ij} which is necessary to measure distances on the manifold. We see this as a necessary first step towards a description of a fluid in general relativity which requires to treat time on the same footing as space and therefore it is necessary to introduce a g_{00} component of the more general metric tensor $g_{\mu\nu}$ [6–8].

Aim of statistical physics is to understand whether the architecture of the statistical mechanics theory allows to reproduce experimental results through the study of a particular model of a fluid. The most common case is that of a fluid living in the Euclidean flat space [9]. Recently the interest moved from the fluid to the Riemannian manifold where the fluid lives, which then becomes the protagonist of the statistical mechanics problem. An example of this change of point of view is offered by some results for the classical two dimensional Coulomb plasma at a special value of the Coupling constant allowing an analytic exact solution for the partition function on various surfaces: the plane [10], the cylinder [11], the sphere [12], the pseudosphere [13], the Flamm paraboloid [14, 15].

In a recent work [16] we studied a quantum fluid on a sphere through the path integral Monte Carlo method. We left as an open problem the calculation of the pressure. In Ref. [13] we studied the pressures of a classical Coulomb plasma on a pseudosphere. In particular we proved the consistency of the virial and kinetic routes to the pressures.

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In this work we will consider the problem of determining a pressure estimator that could be used in a path integral calculation of the pressure of a quantum fluid on a Riemannian manifold. We will propose three routes: virial, thermodynamic, and kinetic. Discuss the consistency of one route with the other and the requirement of having the correct quantum Euclidean limit and classical Riemannian limit.

II. CANONICAL VARIABLES

Consider a Riemannian manifold of dimension d. We will use a bold symbol to denote the d dimensional point on the manifold. We will also adopt Einstein summation convention to tacitly assume a summation over repeated indexes.

The distance between two neighboring points on the manifold is

$$ds^2 = g_{ij}(\boldsymbol{q})\, dq^i dq^j, \tag{2.1}$$

therefore the volume element on the manifold is

$$d\mu = \sqrt{g}\, d\boldsymbol{q}, \tag{2.2}$$

where $g = \det||g_{ij}||$.

A description of a classical fluid on a Riemannian manifold requires the introduction of a Hamiltonian $H(\boldsymbol{Q}, \boldsymbol{P}) = T(\boldsymbol{Q}, \boldsymbol{P}) + V(\boldsymbol{Q})$ where $\boldsymbol{Q} = (\boldsymbol{q}_1, \boldsymbol{q}_2, \dots, \boldsymbol{q}_N)$ are the coordinates of the N particles, $\boldsymbol{P} = (\boldsymbol{p}_1, \boldsymbol{p}_2, \dots, \boldsymbol{p}_N)$ are the momenta,

$$T = \frac{1}{2m} \tilde{g}^{ij}(\boldsymbol{Q})\, P_i P_j, \tag{2.3}$$

is the kinetic energy of the particles of mass m, and V is their potential energy. In Eq. (2.3) we introduced the metric tensor in the manifold of dimension Nd representing the configuration space of the N particles, i.e.

$$\tilde{g}_{ij}(\boldsymbol{Q}) = g_{i_1 j_1}(\boldsymbol{q}_1) \otimes \dots \otimes g_{i_N j_N}(\boldsymbol{q}_N), \tag{2.4}$$

$$\tilde{g}(\boldsymbol{Q}) = \prod_{\mu=1}^{N} g(\boldsymbol{q}_\mu), \tag{2.5}$$

where $||\tilde{g}_{ij}||$ is a matrix made of N diagonal blocks $||g_{i_\mu j_\mu}||$ with $\mu = 1, 2, \dots, N$.

Quantization can be obtained in several ways [17]. The simplest way, in position representation, is to let $\hat{q}^i = q^i$ be just a multiplicative coordinate and $\hat{p}_i = -i\hbar[\partial_i + A_i(\boldsymbol{q})]$ with $\partial_i = \partial/\partial q^i$ the momentum operator. Then it is a simple exercise to prove that in order to have a self adjoint momentum operator respect to the volume measure $d\mu$ it is necessary to choose

$$A_i(\boldsymbol{q}) = \frac{1}{2}\partial_i \ln \sqrt{g(\boldsymbol{q})}. \tag{2.6}$$

We can then propose various different quantizations of the kinetic energy depending on how we order the momenta and the metric tensor in the classical expression (2.3) [1, 18]. A choice satisfying Hermiticity is the following

$$\hat{T} = \frac{1}{2m} \hat{P}_i \tilde{g}^{ij}(\boldsymbol{Q}) \hat{P}_j = -\frac{\hbar^2}{2m} \tilde{g}^{-1/4} \Delta\, \tilde{g}^{1/4}, \tag{2.7}$$

where $\Delta = \partial_i \tilde{g}^{ij} \partial_j$. It can be proven [19] that

$$\tilde{g}^{-1/4} \Delta\, \tilde{g}^{1/4} = \Delta_{\rm LB} - v_{\rm eff}, \tag{2.8}$$

where $\Delta_{\rm LB}$ is the Laplace-Beltrami operator

$$\Delta_{\rm LB} = \tilde{g}^{-1/2} \partial_i \tilde{g}^{1/2} \tilde{g}^{ij} \partial_j, \tag{2.9}$$

and

$$v_{\rm eff} = \tilde{g}^{1/4} \Delta_{\rm LB} \tilde{g}^{-1/4}, \tag{2.10}$$

where in this expression the derivatives stop acting on the last function. In Riemann normal coordinates at a given point, one has

$$v_{\rm eff} = R/6, \tag{2.11}$$

where R is the Ricci scalar curvature. It is important to stress that in this expression v_{eff} depends on the coordinate representation of the metric determinant, whereas the Ricci scalar is a coordinate invariant geometric scalar.

We find this way of quantizing the kinetic energy particularly elegant since we recover the Laplace-Beltrami operator which is obtained from the Laplacian in a tangent plane by substituting partial derivatives with covariant derivatives. This makes the Laplace-Beltrami operator the natural choice as a result of quantization of free particles on the Reimannian manifold. Whereas v_{eff} is just a function of the coordinates $\boldsymbol{Q}$ which depends on the curvature of the manifold and in the Euclidean limit $g_{ij} \to \delta_{ij}$ with δ_{ij} the Kronecker delta, g is a constant and $v_{\text{eff}} \to 0$.

In the next section following DeWitt [1] we will determine a position representation of a density matrix $\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \beta)$ describing the thermal equilibrium of the fluid at an inverse temperature $\beta = 1/k_B T$, with k_B Boltzmann constant and T absolute temperature, satisfying to Block equation

$$\frac{\partial}{\partial \beta}\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \beta) = -\hat{H}\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \beta), \tag{2.12}$$

$$\rho(\boldsymbol{Q}, \boldsymbol{Q}'; 0) = \delta(\boldsymbol{Q} - \boldsymbol{Q}')/\sqrt{\tilde{g}}. \tag{2.13}$$

where $\hat{T} = -\lambda \Delta_{\text{LB}}$ acts on $\boldsymbol{Q}$ and $\lambda = \hbar^2/2m$. We choose this kinetic energy neglecting v_{eff} because the Laplace-Beltrami operator is the most natural choice. We will express the density matrix as a path integral on the manifold. We will discretize the path at equispaced inverse temperatures $k\tau$ with $k = 0, 1, 2, \ldots, M$ such that $M\tau = \beta$. And we will give the explicit expression of the small inverse temperature density matrix $\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \tau)$.

III. PRIMITIVE APPROXIMATION FOR THE DENSITY MATRIX

DeWitt wrote a generalized, coordinate invariant ansatz for the small inverse temperature density matrix between two nearby points $\boldsymbol{Q}$ and $\boldsymbol{Q}'$

$$\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \tau) \propto \tilde{g}(\boldsymbol{Q})^{-1/4} \sqrt{\mathscr{D}(\boldsymbol{Q}, \boldsymbol{Q}'; \tau)}\, \tilde{g}(\boldsymbol{Q}')^{-1/4} e^{-\mathscr{S}(\boldsymbol{Q}, \boldsymbol{Q}'; \tau)}, \tag{3.1}$$

$$\mathscr{S}(\boldsymbol{Q}, \boldsymbol{Q}'; \tau) = d\tilde{s}^2(\boldsymbol{Q}, \boldsymbol{Q}')/4\lambda\tau + \tau V(\boldsymbol{Q}), \tag{3.2}$$

$$d\tilde{s}^2(\boldsymbol{Q}, \boldsymbol{Q}') = \tilde{g}_{ij}(\boldsymbol{Q})(Q^i - Q'^i)(Q^j - Q'^j). \tag{3.3}$$

Here

$$\mathscr{D}_{ij} = -\partial_i \partial'_j \mathscr{S}(\boldsymbol{Q}, \boldsymbol{Q}'; \tau), \tag{3.4}$$

$$\mathscr{D}(\boldsymbol{Q}, \boldsymbol{Q}'; \tau) = \det ||\mathscr{D}_{ij}||, \tag{3.5}$$

is the van Vleck determinant [2, 20]. This determinant is the Jacobian of the transformation from the initial conditions given by fixing the pair of momentum and coordinate of each particle to the boundary conditions given by specifying the pair of initial and final coordinates needed in the coordinate representation. The square root of the van Vleck determinant factor takes into account the density of paths among the minimum extremal region for the action (see Chapter 12 of Ref. [2]).

For the path integral to be valid, this small inverse temperature density matrix must act as a kernel that evolves the density matrix according to the covariant Block equation to first order in τ:

$$\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \beta + \tau) = \int \rho(\boldsymbol{Q}, \boldsymbol{Q}''; \tau)\rho(\boldsymbol{Q}'', \boldsymbol{Q}'; \beta)\sqrt{\tilde{g}(\boldsymbol{Q}'')}\, d\boldsymbol{Q}'', \tag{3.6}$$

To evaluate the integral, DeWitt set the origin of a Riemann Normal Coordinate system at the final point $\boldsymbol{Q}$, treating the initial point $\boldsymbol{Q}''$ as a small displacement $\boldsymbol{\xi} = \boldsymbol{Q}'' - \boldsymbol{Q}$. Expanding $\mathscr{S}$, $\mathscr{D}$, $\sqrt{\tilde{g}}$, and ρ for $\boldsymbol{Q}''$ around $\boldsymbol{Q}$, performing the Gaussian integrals in $\boldsymbol{\xi}$ and keeping only first order terms in τ one finds

$$\frac{\partial}{\partial \beta}\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \beta) = -\left[\hat{H} - \frac{\hbar^2}{2m}\eta R(\boldsymbol{Q})\right]\rho(\boldsymbol{Q}, \boldsymbol{Q}'; \beta), \tag{3.7}$$

where $\hat{H} = -\lambda\Delta_{\text{LB}} + V(\boldsymbol{Q})$, $R(\boldsymbol{Q})$ is the Ricci scalar curvature, and η is a coefficient which depends on the chosen inverse temperature slicing prescription such as prepoint, midpoint, or postpoint in the evaluation of the metric tensor in (3.3). DeWitt concluded that the small inverse temperature action $\mathscr{S}$ cannot simply be the classical action. It must include an effective quantum potential to subtract the discrepancy. Block equation (3.7) is in agreement with the result (2.11) upon choosing $\eta = 1/6$. This makes sense because in the path integral you work on a tangent plane at each imaginary time slice.

IV. ROUTES TO THE PRESSURE

In Ref. [13] are described 3 different routes to the pressure of a classical fluid in a Riemannian manifold (CFR): kinetic, thermodynamic, and virial. It is then proven the consistency between the kinetic and the virial pressure estimators but not with the thermodynamic one

In Ref. [9] are described 2 different routes to the pressure of a quantum fluid in Euclidean space (QFE): thermodynamic and virial. It is then proven the consistency between the two pressure estimators.

Here we will extend the three estimators to a quantum fluid on a Riemannian manifold (QFR) and discuss their consistency and whether they have the correct CFR and QFE limits.

A. Virial route

We consider the fluid as confined in a region Ω of the Riemannian manifold by an external potential $V_{\rm ext}(\boldsymbol{Q})$. So that $V(\boldsymbol{Q}) = V_{\rm int}(\boldsymbol{Q}) + V_{\rm ext}(\boldsymbol{Q})$ with $V_{\rm int}$ the internal potential due to the interactions of the fluid particles.

Consider next the operator

$$\hat{O} = \frac{1}{2}\left(\hat{Q}^i\hat{P}_i + \hat{P}^i\hat{Q}_i\right), \tag{4.1}$$

and its time derivative

$$\frac{d\hat{O}}{dt} = \frac{i}{\hbar}\left[\hat{H},\hat{O}\right], \tag{4.2}$$

where $[\ldots]$ denotes the commutator.

The thermal average of $\hat{O}$

$$\langle\hat{O}\rangle_\rho = \frac{{\rm tr}(\hat{\rho}\hat{O})}{{\rm tr}\hat{\rho}}, \tag{4.3}$$

with $\hat{\rho} = \exp(-\beta\hat{H})$ the density matrix and ${\rm tr}(\ldots)$ denotes the trace, will remain constant in time at thermal equilibrium. So that

$$\left\langle\frac{d\hat{O}}{dt}\right\rangle_\rho = \frac{d\langle\hat{O}\rangle_\rho}{dt} = 0. \tag{4.4}$$

Considering the following elementary commutators

$$\left[\hat{Q}^i,\hat{Q}^j\right] = 0, \tag{4.5}$$

$$\left[\hat{P}_i,\hat{P}_j\right] = -\hbar^2\left\{[\partial_i,A_j] + [A_i,\partial_j]\right\}, \tag{4.6}$$

$$= -\hbar^2\left\{\partial_i A_j - \partial_j A_i\right\} = 0, \tag{4.7}$$

$$\left[\hat{P}_i,\hat{Q}^j\right] = -i\hbar\delta_{ij}, \tag{4.8}$$

$$\left[\hat{P}_i,\tilde{g}^{jk}\right] = -i\hbar\partial_i\tilde{g}^{jk}, \tag{4.9}$$

where we used the commutation of partial derivatives, we find

$$\left[\hat{V},\hat{O}\right] = 2i\hbar Q^i\partial_i V, \tag{4.10}$$

$$\left[\hat{T},\hat{O}\right] = -2i\hbar(\hat{T} + \hat{P}_i\hat{P}_j\tilde{g}^{ij}) + i\hbar[Q^k\hat{P}_i(\partial_k\tilde{g}^{ij})\hat{P}_j + \hat{P}_i(\partial_k\tilde{g}^{ij})\hat{P}_jQ^k]. \tag{4.11}$$

From Eqs. (4.2) and (4.4) follows

$$\left\langle\left[\hat{T},\hat{O}\right]\right\rangle_\rho = -\left\langle\left[\hat{V},\hat{O}\right]\right\rangle_\rho = -i\hbar\left\langle Q^i\partial_i V\right\rangle_\rho = -i\hbar\left\langle Q^i\partial_i V_{\rm int}\right\rangle_\rho - i\hbar P_v\int_{\partial\Omega} q^i\,dS_i, \tag{4.12}$$

where the first term on the right hand side is the virial of the forces and the second term is due to the virial pressure P_v exerted by the fluid particles on the walls at the boundary $\partial\Omega$ of the region containing it. By Gauss theorem

$$\int_{\partial\Omega} q^i \, dS_i = \int_\Omega \partial_i \left(\sqrt{g} q^i\right) \, d\boldsymbol{q}. \tag{4.13}$$

In the Euclidean limit $g_{ij} \to \delta_{ij}$, $[\hat{T}, \hat{O}]i/\hbar \to \hat{P}_i \hat{P}_i/m$ and we recover the correct QFE limit [4]

$$3P_v V_\Omega = 2\left\langle \hat{T} \right\rangle_\rho - \left\langle \hat{W} \right\rangle_\rho, \tag{4.14}$$

where V_Ω is the volume of the region Ω and

$$W = Q_i \partial_i V_{\rm int}, \tag{4.15}$$

is the virial of the forces.

In the classical limit on a Riemannian manifold (CFR) all the operators become numbers and we recover Eq. (3.7) of Ref. [13].

So our estimator of Eq. (4.12) for the virial pressure of the QFR case has both the correct CFR limit and the QFE limit.

B. Thermodynamic pressure

The link between statistical mechanics and thermodynamics is established by the relation

$$\beta A = -\ln Z, \tag{4.16}$$

where $Z = {\rm tr}\hat{\rho}$, with $\hat{\rho} = e^{-\beta \hat{H}}$, is the partition function and A is the Helmholtz free energy that is the thermodynamic potential for a system of fixed N, $V_\Omega = \int_\Omega \sqrt{g}\, d\boldsymbol{q}$, and T. If we start from the defining relation

$$A = U - k_B T S, \tag{4.17}$$

where U is the internal energy and S is the entropy. From the first principle of thermodynamics [21] it is a simple exercise to prove that

$$P_T = -\left(\frac{\partial A}{\partial V_\Omega}\right)_{\beta,N}, \tag{4.18}$$

$$S = -\left(\frac{\partial A}{\partial k_B T}\right)_{V_\Omega,N}, \tag{4.19}$$

$$U = \left(\frac{\partial \beta A}{\partial \beta}\right)_{V_\Omega,N}. \tag{4.20}$$

We then see immediately that the internal energy (4.20) gives the desired direct estimator $U = \langle \hat{H} \rangle_\rho = {\rm tr}(\hat{\rho}\hat{H})/{\rm tr}\hat{\rho}$ and the entropy (4.19) the desired direct estimator $S = \beta(U - A) = -{\rm tr}[\hat{\rho}\ln(\hat{\rho}/Z)]/{\rm tr}\hat{\rho}$.

In the QFE case the thermodynamic estimator for the pressure (4.35) can be reached [9] through a coordinates scaling

$$Q_i \to \xi Q_i, \tag{4.21}$$

$$V_\Omega = L^d \to \xi^d L^d, \tag{4.22}$$

where ξ is the scaling parameter and we chose the region Ω containing the fluid as an hypercube of side L. Then

$$P_T = -\frac{1}{d\Omega}\left(\frac{\partial A}{\partial \xi}\right)_{\xi=1,\beta,N}. \tag{4.23}$$

In extending to the quantum Riemannian QFR case

$$\tilde{g}_{ij}(\boldsymbol{Q}) \to \tilde{g}_{ij}(\xi \boldsymbol{Q}), \tag{4.24}$$

$$\left.\frac{\partial \tilde{g}_{ij}(\xi \boldsymbol{Q})}{\partial \xi}\right|_{\xi=1} = Q^k \partial_k \tilde{g}_{ij}(\boldsymbol{Q}) = Q^k \tilde{g}_{ij,k}(\boldsymbol{Q}), \tag{4.25}$$

where we introduced the comma notation to denote a partial derivative. For the DeWitt action (3.2)-(3.3) we then find

$$\left.\frac{\partial \mathscr{S}}{\partial \xi}\right|_{\xi=1} = \tilde{g}_{ij}(\boldsymbol{Q})\frac{(\boldsymbol{Q}-\boldsymbol{Q}')^i(\boldsymbol{Q}-\boldsymbol{Q}')^j}{2\lambda\tau} + \tag{4.26}$$

$$Q^k\tilde{g}_{ij,k}(\boldsymbol{Q})\frac{(\boldsymbol{Q}-\boldsymbol{Q}')^i(\boldsymbol{Q}-\boldsymbol{Q}')^j}{4\lambda\tau} + \tag{4.27}$$

$$\tau Q^i V_{,i}(\boldsymbol{Q}), \tag{4.28}$$

where we recognize in the last term the virial of the forces already introduced in Eq. (4.15). For the volume of the region Ω we find

$$\left.\frac{\partial V_\Omega}{\partial \xi}\right|_{\xi=1} = \int_\Omega \left[d + Q^i\partial_i\sqrt{g}\right]\, d\boldsymbol{q} \tag{4.29}$$

where $\partial_i\sqrt{g} = \sqrt{g}g^{jk}g_{jk,i} = \sqrt{g}\Gamma^j{}_{ji}$ vanishes in the Euclidean limit and $\partial V_\Omega/\partial\xi|_{\xi=1} = dV_\Omega$. And for the Ricci scalar

$$\left.\frac{\partial R(\boldsymbol{Q})}{\partial \xi}\right|_{\xi=1} = Q^i R_{,i}(\boldsymbol{Q}). \tag{4.30}$$

We then reach the following expression for the thermodynamic estimator of the pressure

$$\beta P_T = \left.\frac{\partial\xi}{\partial V_\Omega}\frac{\partial \ln Z}{\partial\xi}\right|_{\xi=1} \tag{4.31}$$

$$= \left.\frac{\partial\xi}{\partial V_\Omega}\frac{1}{Z}\frac{\partial}{\partial\xi}\int \rho(\boldsymbol{Q},\boldsymbol{Q};\beta)\sqrt{\tilde{g}(\boldsymbol{Q})}\, d\boldsymbol{Q}\right|_{\xi=1} \tag{4.32}$$

$$= \left.\frac{\partial\xi}{\partial V_\Omega}\frac{1}{Z}\frac{\partial}{\partial\xi}\int d\mu\, \rho(\boldsymbol{Q},\boldsymbol{Q}_1;\tau)\rho(\boldsymbol{Q}_1,\boldsymbol{Q}_2;\tau)\cdots\rho(\boldsymbol{Q}_{M-1},\boldsymbol{Q};\tau)\right|_{\xi=1}, \tag{4.33}$$

where in the last step we expanded the position representation of the density matrix into a pat integral, discretized the path at equispaced inverse temperatures $k\tau$ with $k = 1, 2, \ldots, M-1$ such that $M\tau = \beta$, and used the DeWitt primitive approximation (3.1) for the small inverse temperature τ of the density matrix $\rho(\boldsymbol{Q}_k, \boldsymbol{Q}_{k+1};\tau)$ which constitutes the kth link. The integration measure for the path integral is

$$d\mu = \sqrt{\tilde{g}(\boldsymbol{Q})}\, d\boldsymbol{Q}\prod_{k=1}^{M-1}\sqrt{\tilde{g}(\boldsymbol{Q}_k)}\, d\boldsymbol{Q}_k. \tag{4.34}$$

Then, when calculating the ξ derivative in the path integral of Eq. (4.33) we can act on the kth link, each link contributing an equal amount to the thermal average,

$$\beta P_T = \left.\frac{\partial\xi}{\partial V_\Omega}\left\{M\left\langle -\frac{\partial\mathscr{S}_k}{\partial\xi} + \tau\lambda\eta\frac{\partial R(\boldsymbol{Q}_k)}{\partial\xi} + \alpha_k\right\rangle_\rho\right\}\right|_{\xi=1}, \tag{4.35}$$

where $\mathscr{S}_k$ is the action relative to the kth link and α_k is the contribution due to the ξ derivative of the measure $d\mu$ relative to the kth link

$$\alpha_k = dN + \frac{Q_k^i\left[\sqrt{\tilde{g}(\boldsymbol{Q}_k)}\right]_{,i}}{\sqrt{\tilde{g}(\boldsymbol{Q}_k)}}. \tag{4.36}$$

We immediately see that in the Euclidean limit $g_{ij} \to \delta_{ij}$

$$P_T = \frac{1}{dV_\Omega\tau}\left\langle dN - \frac{(\boldsymbol{Q}_k-\boldsymbol{Q}_{k+1})^2}{2\lambda\tau} - \tau\boldsymbol{Q}_k\cdot\boldsymbol{\nabla}_k V(\boldsymbol{Q}_k)\right\rangle_\rho, \tag{4.37}$$

and we recover the result of Section VI B of Ref. [9] which in turn coincides with the virial estimator in the same QFE case [9].

Although we recover the desired QFE limit there is no consistency between the thermodynamic (4.37) and the virial (4.14) routes in the full QFR case. In fact the virial route requires second derivatives of the metric tensor whereas the thermodynamic route only first ones.

It is important to stress that the thermodynamic estimator for the pressure (4.35) does not reduce to the virial estimator in the CFR limit of Ref. [13]. In fact taking the limit $\tau \to \beta$ in Eq. (4.35) already the term $\partial V_\Omega/\partial\xi|_{\xi=1}$ does not appear in the CFR limit of Eq. (4.14).

Things get even worst when extending our results to a fluid in general relativity [22–26]. In this case we can use the 3+1 Arnowitt, Deser, and Misner foliation of space and imaginary time [27]. It is also necessary to treat the thermal energy ϵ as $\epsilon^2 = \tilde{g}^{00}(k_B T)^2$ which requires to introduce a Tolman-Eherenfest temperature [28] $\beta_{\rm TE}(R) = M\tau_{\rm TE}(\boldsymbol{R}) = \beta\sqrt{\tilde{g}_{00}(R)}$ which is a function of space $\boldsymbol{Q}$ and imaginary time Q^0 with $R = (Q^0, \boldsymbol{Q})$ and the elementary distance between two neighboring events $ds^2 = g_{\mu\nu}dr^\mu dr^\nu$ where now the Greek indexes run over the 4 components of space and imaginary time.

C. Kinetic pressure

In Ref. [13] a kinetic route to the pressure of a classical fluid on a Riemannian manifold was presented and proven to be equivalent to the thermodynamic route.

Here we will discuss how such route could be extended to the quantum regime. Consider then the one body density [9]

$$n(\boldsymbol{q}, \boldsymbol{q}') = \frac{N}{Z}\int \rho(\boldsymbol{q}, \boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_N, \boldsymbol{q}', \boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_N; \beta) \prod_{i=2}^{N} \sqrt{g(\boldsymbol{q}_i)} d\boldsymbol{q}_i, \tag{4.38}$$

where the non diagonal density matrix in the integrand is built from open paths. Then $\int n(\boldsymbol{q}, \boldsymbol{q})\sqrt{g(\boldsymbol{q})}d\boldsymbol{q} = N$. From the one body density we can extract the kinetic energy

$$\left\langle \hat{T} \right\rangle_\rho = -\lambda \int_\Omega \Delta_{\rm LB} n(\boldsymbol{q}, \boldsymbol{q}')|_{\boldsymbol{q}'=\boldsymbol{q}} \sqrt{g(\boldsymbol{q})} d\boldsymbol{q}, \tag{4.39}$$

which in the Euclidean limit $g_{ij} \to \delta_{ij}$ becomes [9]

$$\left\langle \hat{T} \right\rangle_\rho \to \left\langle \frac{dN}{2\tau} - \frac{(\boldsymbol{Q}_k - \boldsymbol{Q}_{k+1})^2}{4\lambda\tau^2} \right\rangle_\rho. \tag{4.40}$$

Then for the ideal gas we could define the following kinetic estimator

$$P_K = \frac{2}{dV_\Omega}\left\langle \hat{T} \right\rangle_\rho, \tag{4.41}$$

which has the correct Euclidean limit (4.37). In the interacting fluid we then need to add the virial of the forces.

Although we recover the desired QFE limit there is no consistency between the thermodynamic (4.35) or the virial (4.14) routes and the kinetic route in the full QFR case.

V. CONCLUSIONS

We proposed three estimators for the pressure of a quantum fluid on a Riemannian manifold (QFR) that are reached following the virial, thermodynamic, and kinetic routes. Although we could not find consistency between the three estimators we proved that the virial has the correct Euclidean limit (QFE) and classical limit (CFR), the thermodynamic and the kinetic has only the correct Euclidean limit.

The inconsistency between our virial and thermodynamic routes has to be looked into the problem of operators ordering chosen in the operator $\hat{O}$ in Eq. (4.1). On the other hand, the inconsistency between the kinetic and the thermodynamic routes appears already in how the volume of the confining region enters in the two expressions.

We therefore suggest our virial route as the best candidate for a pressure estimator. As long as the consistency between the virial and the thermodynamic routes remains unsettled.

The treatment of the fluid in general relativity requires the use of the Tolman-Eherenfest temperature. This will affect the thermodynamic estimator but not the virial or the kinetic ones.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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